

Recursion Function in Python - 30 Examples with Code and Output

1. Factorial of a Number

```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)

print(factorial(5))
```

Output: 120

2. Sum of First N Natural Numbers

```
def sum_n(n):
    if n == 0:
        return 0
    return n + sum_n(n - 1)

print(sum_n(5))
```

Output: 15

3. Nth Fibonacci Number

```
def fibonacci(n):
    if n <= 1:
        return n
    return fibonacci(n - 1) + fibonacci(n - 2)

print(fibonacci(6))
```

Output: 8

4. Print Numbers from 1 to N

```
def print_1_to_n(n):
    if n == 0:
        return
    print_1_to_n(n - 1)
    print(n)

print_1_to_n(5)
```

Output: 1 2 3 4 5

5. Print Numbers from N to 1

```
def print_n_to_1(n):
    if n == 0:
        return
    print(n)
    print_n_to_1(n - 1)

print_n_to_1(5)
```

Output: 5 4 3 2 1

6. Sum of Digits

```
def sum_digits(n):
    if n == 0:
```

```

        return 0
    return n % 10 + sum_digits(n // 10)

print(sum_digits(1234))

```

Output: 10

7. Product of Digits

```

def product_digits(n):
    if n == 0:
        return 1
    return (n % 10) * product_digits(n // 10)

print(product_digits(234))

```

Output: 24

8. Reverse a Number

```

def reverse_number(n, rev=0):
    if n == 0:
        return rev
    return reverse_number(n // 10, rev * 10 + n % 10)

print(reverse_number(1234))

```

Output: 4321

9. Check Number Palindrome

```

def is_palindrome(n, temp=None):
    if temp is None:
        temp = n
    def rev(num, r=0):
        if num == 0:
            return r
        return rev(num // 10, r * 10 + num % 10)
    return n == rev(temp)

print(is_palindrome(121))

```

Output: True

10. Count Digits in a Number

```

def count_digits(n):
    if n == 0:
        return 0
    return 1 + count_digits(n // 10)

print(count_digits(12345))

```

Output: 5

11. Reverse a String

```

def reverse_string(s):
    if len(s) == 0:
        return s
    return s[-1] + reverse_string(s[:-1])

print(reverse_string('hello'))

```

Output: olleh

12. String Palindrome

```

def is_palindrome_str(s):

```

```

    if len(s) <= 1:
        return True
    if s[0] != s[-1]:
        return False
    return is_palindrome_str(s[1:-1])

```

```
print(is_palindrome_str('madam'))
```

Output: True

13. Count Vowels in String

```

def count_vowels(s):
    if s == '':
        return 0
    return (1 if s[0].lower() in 'aeiou' else 0) + count_vowels(s[1:])

```

```
print(count_vowels('Hello'))
```

Output: 2

14. Replace Character in String

```

def replace_char(s, old, new):
    if not s:
        return ''
    ch = new if s[0] == old else s[0]
    return ch + replace_char(s[1:], old, new)

```

```
print(replace_char('apple', 'p', 'b'))
```

Output: abble

15. Remove Spaces

```

def remove_space(s):
    if s == '':
        return ''
    if s[0] == ' ':
        return remove_space(s[1:])
    return s[0] + remove_space(s[1:])

```

```
print(remove_space('a b c'))
```

Output: abc

16. Max Element in List

```

def find_max(lst):
    if len(lst) == 1:
        return lst[0]
    return max(lst[0], find_max(lst[1:]))

```

```
print(find_max([1, 4, 7, 2]))
```

Output: 7

17. Sum of List Elements

```

def sum_list(lst):
    if not lst:
        return 0
    return lst[0] + sum_list(lst[1:])

```

```
print(sum_list([1,2,3,4]))
```

Output: 10

18. Count Even Numbers

```
def count_even(lst):
    if not lst:
        return 0
    return (1 if lst[0] % 2 == 0 else 0) + count_even(lst[1:])

print(count_even([1,2,3,4,6]))
```

Output: 3

19. Linear Search

```
def search(lst, x):
    if not lst:
        return False
    if lst[0] == x:
        return True
    return search(lst[1:], x)

print(search([1,2,3,4], 3))
```

Output: True

20. Binary Search

```
def binary_search(lst, low, high, x):
    if low > high:
        return False
    mid = (low + high) // 2
    if lst[mid] == x:
        return True
    elif lst[mid] > x:
        return binary_search(lst, low, mid - 1, x)
    else:
        return binary_search(lst, mid + 1, high, x)

print(binary_search([1,2,3,4,5], 0, 4, 3))
```

Output: True

21. GCD of Two Numbers

```
def gcd(a, b):
    if b == 0:
        return a
    return gcd(b, a % b)

print(gcd(12, 18))
```

Output: 6

22. LCM using GCD

```
def gcd(a, b):
    if b == 0:
        return a
    return gcd(b, a % b)

def lcm(a, b):
    return (a * b) // gcd(a, b)

print(lcm(12, 18))
```

Output: 36

23. Power Function

```
def power(x, n):
    if n == 0:
        return 1
    return x * power(x, n - 1)

print(power(2, 5))
```

Output: 32

24. Triangular Number

```
def triangular(n):
    if n == 0:
        return 0
    return n + triangular(n - 1)

print(triangular(5))
```

Output: 15

25. Sum of Even Numbers

```
def sum_even(n):
    if n == 0:
        return 0
    if n % 2 == 0:
        return n + sum_even(n - 2)
    else:
        return sum_even(n - 1)

print(sum_even(10))
```

Output: 30

26. Print String N Times

```
def print_n_times(s, n):
    if n == 0:
        return
    print(s)
    print_n_times(s, n - 1)

print_n_times('Hi', 3)
```

Output: Hi Hi Hi

27. Print List in Reverse

```
def print_reverse(lst):
    if not lst:
        return
    print(lst[-1])
    print_reverse(lst[:-1])

print_reverse([1, 2, 3])
```

Output: 3 2 1

28. Decimal to Binary

```
def dec_to_bin(n):
    if n == 0:
        return ''
    return dec_to_bin(n // 2) + str(n % 2)

print(dec_to_bin(10))
```

Output: 1010

29. Count Zeroes in Number

```
def count_zero(n):
    if n == 0:
        return 0
    return (1 if n % 10 == 0 else 0) + count_zero(n // 10)
```

```
print(count_zero(10020))
```

Output: 3

30. Tower of Hanoi

```
def tower_of_hanoi(n, source, target, auxiliary):  
    if n == 1:  
        print(f"Move disk 1 from {source} to {target}")  
        return  
    tower_of_hanoi(n-1, source, auxiliary, target)  
    print(f"Move disk {n} from {source} to {target}")  
    tower_of_hanoi(n-1, auxiliary, target, source)
```

```
tower_of_hanoi(3, 'A', 'C', 'B')
```

Output: Move disk 1 from A to C Move disk 2 from A to B Move disk 1 from C to B Move disk 3 from A to C Move disk 1 from B to A Move disk 2 from B to C Move disk 1 from A to C